

Polynomial Regression Assignment — Report

Roll No. **BT2024154** | Problems: var1 (steam turbine Net Power Score), var2 (thermal anomaly mapping)

Code repository: <https://github.com/Kkshiteej-Tiwari/Machine-Learning-Assignment-1>

1. Summary

Both problems were solved with **polynomial regression only**: every model is a polynomial in the inputs containing all terms of total degree $\leq d$, fitted by penalised least squares. The degree and the penalty were chosen by repeated 5-fold cross-validation (CV). The final choices are **degree 5** for var1 (CV MSE 0.351, CV R^2 0.967) and **degree 8** for var2 (CV MSE 0.249, CV R^2 0.9945). The largest allowed degrees (10 and 20) overfit badly. For var2 the CV error flattens at about 0.25 for degrees 8–10, which looks like the noise floor, so degree 8 is the smallest degree that reaches it.

2. Data exploration

	inputs	train / test rows	Var(y) train	max degree	rows with ≥ 3 inputs at ± 1 (train $\rightarrow$ test)
var1	x1..x6	1000 / 1000	10.49	10	28.7% $\rightarrow$ 66.2%
var2	x1..x3	1000 / 1000	45.52	20	1.5% $\rightarrow$ 1.7%

The files contain no missing values. All inputs lie in $[-1, 1]$, and a large share of values sit *exactly* on ± 1 (31% of var1 training values, 25% for var2). The inputs were evidently clipped to the box during preprocessing. This matters in two ways. (i) High-degree polynomials are least stable near the edges of the domain (Runge-type oscillation), and many points sit on those edges. (ii) For **var1 the test set is shifted toward the corners**: 66% of test rows have three or more clipped inputs, against 29% of training rows. CV on the training rows alone would therefore under-represent the hardest test region. Section 3.3 explains how this is handled. var2 shows no such shift.

3. Approach

3.1 Model family and feature basis

For p inputs and degree d the model is $f(\mathbf{x}) = \sum \mathbf{c}_t \cdot \phi_t(\mathbf{x})$, summed over all exponent vectors $\mathbf{t}$ with $|\mathbf{t}| \leq d$, i.e. $C(p+d, d)$ terms. That is 462 terms for var1 at $d=5$ (8008 at $d=10$) and 165 for var2 at $d=8$ (1771 at $d=20$). This matches the assignment's definition: the powers in each term add up to at most d . Rather than raw monomials $x_1^a x_2^b \dots$, I use the **product-Legendre basis** $P_a(x_1)P_b(x_2)\dots$. Every Legendre product expands into monomials of the same total degree, so it spans *exactly the same set of polynomials* and the model is still plain polynomial regression. The design matrix, however, is far better conditioned on $[-1, 1]$. For var2 the condition number at $d=8$ drops from 1.2×10^3 (monomials) to 24, and at $d=12$ from 7.3×10^4 to 290. This keeps high-degree fits numerically reliable, and it makes a penalty on the coefficients mean the same thing for every term.

3.2 Regularisation

A degree- d polynomial with hundreds or thousands of terms fitted to 800 rows (one CV fold) overfits without a penalty. With $d=6$ for var1 there are more terms (924) than rows, so the training error drops to 0 while the CV error explodes (Figure 1). Two penalised estimators were used:

- **Ridge (L2)**, $\min \|y - \Phi c\|^2 + \alpha \|c\|^2$, with an unpenalised intercept. It is solved through one SVD of the centred design matrix per fold, which gives predictions for all 37 α values (10^{-8} – 10^4) at once. Sweeping every degree (10 for var1, 20 for var2) therefore takes about a minute.
- **Lasso (L1) with a degree-weighted penalty**, $\min \frac{1}{2n} \|y - \Phi c\|^2 + \lambda \sum_t (1+|\mathbf{t}|)^{\gamma/2} |c_t|$. L1 can set coefficients to exactly zero, which helps when the true polynomial uses only a subset of the possible terms. The weight $(1+|\mathbf{t}|)^{\gamma/2}$ penalises high-order terms more than low-order ones, so the fit prefers smooth explanations. It is implemented by rescaling the columns of Φ , and $\gamma=0$ gives ordinary Lasso.

3.3 Model selection

All hyper-parameters (degree d , penalty α or λ , weight γ) were chosen by 5-fold CV, repeated with two different random splits and averaged (seeded, fully reproducible). Stage 1 runs a ridge sweep over *every* degree to map the

bias–variance curve. Stage 2 runs a Lasso grid over the most promising degrees and $\gamma \in \{0,2,4,6\}$. To account for the var1 shift, the selection score is an **importance-weighted CV MSE**. Each validation row with k clipped inputs gets weight $w(k) = P_{\text{test}}(k) / P_{\text{train}}(k)$, so the CV error estimates the error on the test set's mix of rows. Unweighted CV MSE is reported alongside it. For var2 the weights are about 1, and both scores pick the same model.

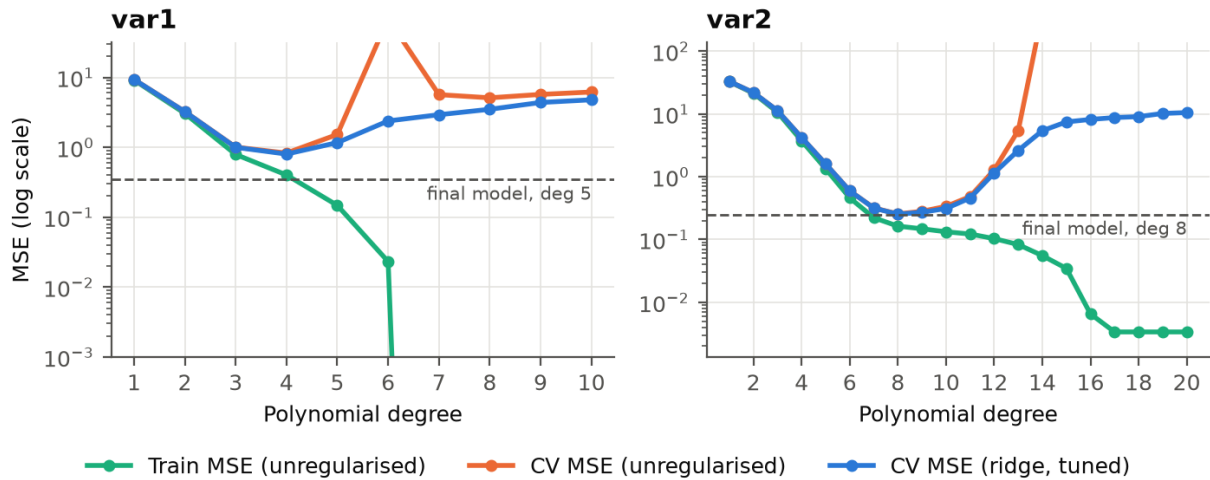


Figure 1. Error against polynomial degree (log scale). Without a penalty the training error keeps falling while the CV error rises sharply past degree 4 (var1) and 9 (var2): the classic overfitting U-curve. Ridge tames the explosion but cannot beat the low-degree optimum. The dashed line is the CV MSE of the final model.

4. Problem var1 — steam turbine Net Power Score (6 inputs, degree ≤ 10)

model	degree	terms (non-zero)	CV MSE	CV R^2	shift-weighted CV MSE
ridge ($\alpha=1e-08$)	2	28	3.236	0.691	5.686
ridge ($\alpha=1e-08$)	3	84	0.999	0.905	1.218
ridge ($\alpha=1e+01$)	4	210	0.800	0.924	0.975
ridge ($\alpha=1e+01$)	5	462	1.166	0.889	1.916
ridge ($\alpha=5e+03$)	6	924	2.383	0.773	2.906
ridge ($\alpha=1e+04$)	10	8008	4.785	0.544	6.710
lasso $\gamma=6$	4		0.619	0.941	0.777
lasso $\gamma=0$	5		0.391	0.963	0.433
lasso $\gamma=6$	6		0.357	0.966	0.388
lasso $\gamma=6$ (final)	5	462 (156)	0.351	0.967	0.381

Chosen degree: 5. Degrees 1–3 clearly underfit: CV MSE falls from 9.34 to 1.00, and train and CV errors stay close together. With ridge the best degree is 4 (CV MSE 0.800), and degree 5 is already worse (1.166). This is because ridge shrinks all 462 coefficients and never removes any, so the extra variance outweighs the gain. The Lasso shows the real structure. The function has genuine degree-5 content but uses only part of the 462 terms: the final model keeps 156 of them, mostly terms that involve one to three inputs. Lasso at degree 5 **more than halves the error** of the best ridge model (0.351 vs 0.800; R^2 0.924 $\rightarrow$ 0.967). Going to degree 6 brings no further gain (0.357), and degrees 7–10 only add variance. The degree-weighted penalty ($\gamma>0$) improves on plain Lasso (0.391 $\rightarrow$ 0.351) because it lets the high-order terms in only when the data supports them, which keeps the surface smooth at the clipped corners the test set is rich in. Degree 5 is also best on the unweighted CV score (lowest 0.350), so the choice of degree does not depend on the shift correction.

Robustness at the corners. The out-of-fold MSE of the final model grows only mildly with the number of clipped inputs (0.27 with none, 0.36 with three, 0.41 with five). The expected test MSE is therefore close to the weighted CV value of about 0.38, not far above it. The final model's test predictions differ from those of the runner-up (degree 6 Lasso) by a mean squared difference of only 0.002, so they are stable.

5. Problem var2 — thermal anomaly mapping (3 inputs, degree ≤ 20)

model	degree	terms (non-zero)	CV MSE	CV R^2	train MSE (no penalty)
ridge ($\alpha=1e+01$)	4	35	4.202	0.9077	3.732

ridge ($\alpha=1e+01$)	6	84	0.594	0.9870	0.460
ridge ($\alpha=5e+00$)	7	120	0.316	0.9931	0.222
ridge ($\alpha=5e+00$)	8	165	0.253	0.9944	0.163
ridge ($\alpha=1e+01$)	9	220	0.271	0.9940	0.148
ridge ($\alpha=5e+00$)	10	286	0.311	0.9932	0.132
ridge ($\alpha=2e+00$)	12	455	1.144	0.9749	0.103
ridge ($\alpha=2e+01$)	14	680	5.395	0.8815	0.055
ridge ($\alpha=2e+03$)	20	1771	10.500	0.7693	0.003
lasso $\gamma=2$ (final)	8	165 (156)	0.249	0.9945	0.166 (penalised)

Chosen degree: 8. The CV curve (Figure 1, right) is a clean U. Error drops steeply up to degree 7–8, stays flat at about 0.25 for degrees 8–10, then rises fast: about 1.1 at degree 12, 5.4 at degree 14 and over 10 at degree 20. The flat bottom suggests the remaining error is mostly measurement noise. The training error at degree 8 (0.163) is only modestly below the CV error (0.253), which fits a noise variance of roughly 0.2. A higher degree cannot reduce that noise; it can only fit it. By the usual parsimony rule (the smallest model whose CV error matches the minimum), degree 8 is chosen. The “up to degree 20” in the brief is an upper bound on the true complexity, not a target. With 1000 samples in 3-D, a degree-20 model has 1771 coefficients, more than the number of points, and even heavy ridge regularisation leaves its CV error 40 times worse.

Ridge and Lasso are practically tied at degree 8 (0.253 vs 0.249). Their test predictions differ by a mean squared difference of only 0.004, so the choice between them is immaterial. The Lasso model was kept because it had the lowest CV score. It zeroes 9 of the 165 terms.

6. Final models and validation

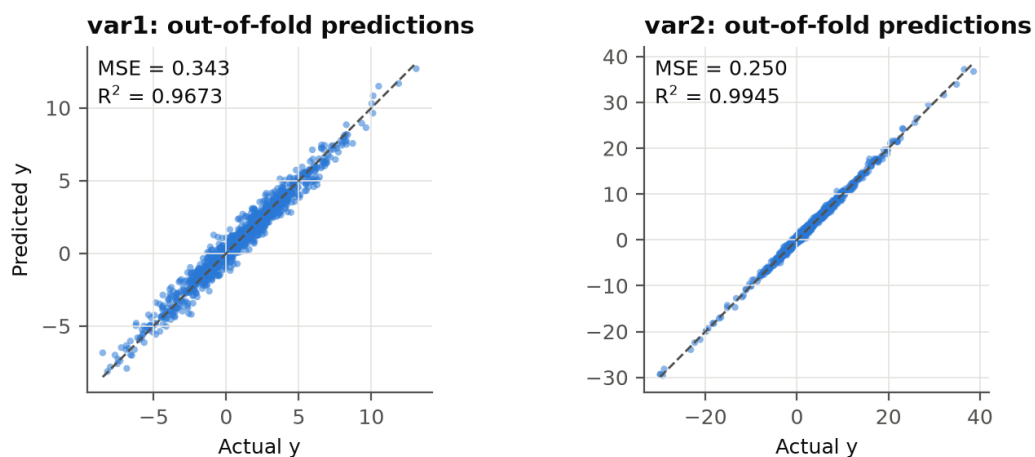


Figure 2. Out-of-fold predictions of the final models against the actual values. No row is predicted by a model that saw it during training.

	degree	method	penalty	non-zero terms	CV MSE	CV R ²	train MSE
var1	5	Lasso, $\gamma=6$	$\lambda=2.2e-04$	156 / 462	0.351	0.9665	0.238
var2	8	Lasso, $\gamma=2$	$\lambda=1.0e-03$	156 / 165	0.249	0.9945	0.166

Each final model was refitted on all 1000 training rows with the selected hyper-parameters and used to predict the test rows. The predictions are in **BT2024154_pred_var1.csv** and **BT2024154_pred_var2.csv**, each a single column y with 1000 rows in the same order as the test files, matching the sample submission.

7. Key decisions at a glance

- **CV over training error** for choosing the degree: training error always favours the highest degree (Figure 1).
- **Legendre basis:** the same polynomial space as monomials, but well conditioned, so degree 8–20 fits are numerically trustworthy.
- **Regularisation:** needed once the number of terms approaches the number of samples. The sparse, degree-weighted Lasso was the decisive improvement for var1.

- **Shift-aware validation:** var1's test set is concentrated at clipped corners, so CV errors were re-weighted to match it, and the chosen model was checked to hold up there.
- **Parsimony:** where several degrees tie within noise (var2: 8–10), the smallest was chosen.

8. Reproducing the results

Repository: <https://github.com/Kkshiteej-Tiwari/Machine-Learning-Assignment-1>. Run `pip install -r requirements.txt`, then `python code/train.py` (model selection and fitting, about 5 minutes on a laptop CPU), then `python code/predict.py` (writes the prediction files). `code/polyreg.py` holds the feature construction, the ridge path, Lasso CV and the final model class. Full CV logs are in `results/`.